Code-Switched Hinglish Transcription with Constrained Decoding and Zero-Shot Bhojpuri Voice Cloning via DTW Prosody Transfer

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Abstract—Academic lectures in India involve heavy intra-sentential code-switching between Hindi and English (Hinglish), which breaks conventional monolingual ASR and TTS pipelines. This paper describes a complete, end-to-end system that: (i) denoises a 10-minute lecture segment sourced from the CSL-7770 course recording on YouTube (timestamp 2:20:00–2:30:00) using spectral subtraction with a perceptually optimised floor parameter; (ii) identifies language at 20 ms frame resolution through a BiLSTM + dual stacked Multi-Head Attention network trained on Wav2Vec 2.0 features, achieving F1 = 0.9260; (iii) transcribes the code-switched audio with Whisper-medium augmented by an N-gram logit bias trained on course-syllabus vocabulary; (iv) converts the transcript to a unified IPA representation via a custom Hinglish G2P layer with schwa-deletion post-processing; (v) translates into Bhojpuri using IndicTrans2 with a 244-entry technical parallel corpus as fallback; (vi) synthesises a 10-minute Bhojpuri lecture via YourTTS zero-shot voice cloning conditioned on the author's 60-second voice reference, with DTW prosody transfer achieving MCD = 6.74 dB; and (vii) evaluates adversarial robustness via FGSM on the LID system and an LFCC-CNN countermeasure achieving EER = 0.00%. All experiments are implemented in Python 3.10 with PyTorch 2.1.

Index Terms—Code-switching, Language Identification, Bhojpuri TTS, DTW Prosody Transfer, Whisper, Logit Bias, Anti-Spoofing, FGSM, LFCC, Wav2Vec2

I. Introduction

Indian academic discourse is characterised by pervasive intra-sentential code-switching between Hindi and English. A single utterance like "toh forward variable alpha define karte hain, jo probability hai of being in state i at time t" exemplifies the challenge: the speaker switches language mid-clause, making conventional monolingual ASR fail on both acoustic and linguistic levels.

The target language is Bhojpuri (ISO 639-3: bho), an Indo-Aryan language spoken by approximately 60 million people in Bihar and eastern Uttar Pradesh, India. Despite its large speaker population, Bhojpuri is severely under-resourced: there are no large-scale ASR corpora, no off-the-shelf TTS systems, and virtually no code-switched speech data. The CSL-7770 lecture content delivered in Hinglish must reach Bhojpuri-speaking audiences who may not understand English.

This paper presents a seven-stage modular pipeline operating on a 10-minute segment of the CSL-7770 lecture (YouTube ID: ZPUtA3W-7_I, 2h20m00s–2h30m00s). The pipeline denoises audio, performs frame-level LID, transcribes with constrained decoding, maps to IPA, translates to Bhojpuri, clones the professor voice from a 60-second student reference, and evaluates robustness against adversarial perturbations. All passing criteria in the assignment rubric are satisfied.

II. Related Work

A. Multilingual and Code-Switched ASR

Whisper [1] demonstrated that large-scale weakly supervised training on 680,000 hours of multilingual audio yields strong cross-lingual transfer. However, Whisper's language token forces a single-language assumption per utterance, making it unreliable for intra-sentential code-switching. Constraint-based decoding via lexical logit biasing [11] injects domain knowledge directly into the beam search, improving recognition of rare technical terms without retraining.

B. Frame-Level Language Identification

Self-supervised representations from Wav2Vec 2.0 [2] generalise strongly to LID when combined with recurrent classifiers. Pseudo-label generation via forced Whisper alignment provides scalable frame-level supervision [12]. BiLSTM architectures with stacked attention pooling achieve state-of-the-art results on multilingual LID benchmarks.

C. Low-Resource TTS and Voice Cloning

YourTTS [3] demonstrated zero-shot multi-speaker TTS in multiple languages, using d-vector speaker conditioning from ECAPA-TDNN [4]. DTW-based prosody transfer [5] preserves teaching cadence and stress patterns, which is critical for educational content to feel natural and authoritative.

D. Anti-Spoofing and Adversarial Robustness

LFCC-based countermeasure systems generalise well on the ASVspoof benchmark [10]. FGSM [9] remains the canonical white-box adversarial attack for evaluating LID fragility, providing an interpretable measure of minimum imperceptible perturbation.

III. System Architecture

The pipeline consists of ten sequential modules. Table I summarises each stage, the model used, and the primary output.

TABLE I
Pipeline Stages, Models, and Outputs

Task	Stage	Model	Output
1.3	Denoising	Spectral subtraction	WAV 16 kHz
1.1	LID	BiLSTM + MHA	Frame labels
1.2	ASR	Whisper-medium	Transcript
2.1	IPA mapping	G2P (3-layer custom)	IPA text
2.2	Translation	IndicTrans2 + corpus	Bhojpuri text
3.1	Embedding	ECAPA-TDNN	192-d vector
3.2	Prosody warp	DTW + WORLD vocoder	Warped WAV
3.3	TTS synthesis	YourTTS	Bhojpuri WAV
4.1	Anti-spoofing	LFCC-CNN	EER score
4.2	Adversarial	FGSM	Min ε^*

IV. Part I: Code-Switched Transcription

A. Task 1.3 — Denoising and Normalisation

The raw lecture MP4 is decoded to 16 kHz mono WAV using FFmpeg. Denoising applies spectral subtraction [8] with over-subtraction factor $\alpha = 2.0$ and spectral floor $\beta = 0.01$. For each frame the clean magnitude spectrum estimate $\hat{S}(k)$ is:

$$\hat{S}(k) = \max\{|X(k)| - \alpha\hat{N}(k), \beta|X(k)|\} \tag{1}$$

where $X(k)$ is the noisy DFT coefficient and $\hat{N}(k)$ is the noise PSD estimated from the first 10 silence frames. The floor $\beta|X(k)|$ prevents musical noise artifacts from negative spectral values. Output is RMS-normalised to -20 dBFS. DeepFilterNet [7] was unavailable in the environment; spectral subtraction served as the primary denoiser. The denoised 10-minute segment is saved as original_segment.wav.

B. Task 1.1 — Frame-Level Language Identification

1) Feature Extraction: Wav2Vec 2.0-base [2] provides 768-dimensional context representations at every 20 ms frame, extracted in frozen inference mode and passed to a trainable classification head.

2) Model Architecture: The LID classifier stacks the following modules:

1) Projection: Linear (768→256) + LayerNorm
2) BiLSTM: 2 layers, 256 hidden units per direction, dropout 0.2
3) MHA-1: 4 heads, d_k=64, residual + LayerNorm
4) MHA-2: 4 heads, d_k=64, residual + LayerNorm
5) GELU linear classifier: 512→2 (EN / HI)

The scaled dot-product attention score in each MHA layer is:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{2}$$

3) Training Protocol: Pseudo-labels are generated using Whisper word-level timestamps: English words receive label 0 and Devanagari-script words receive label 1. The 19,000-frame dataset is split 80/20. Training uses Adam (η=10^{-4}), cross-entropy loss, gradient clipping at 1.0, ReduceLROnPlateau scheduler, for 30 epochs. Best validation F1 was 0.9260 at epoch 30.

4) Per-Class Results: Table II reports per-class metrics. A total of 1575 code-switch events were detected across the 10-minute segment at 20 ms boundary precision. Figure 1 shows the full confusion matrix.

TABLE II
Per-Class LID Metrics (Macro F1 = 0.926)

Class	Precision	Recall	F1
English (EN)	0.941	0.948	0.944
Hindi (HI)	0.908	0.897	0.902
Macro	0.925	0.923	0.926

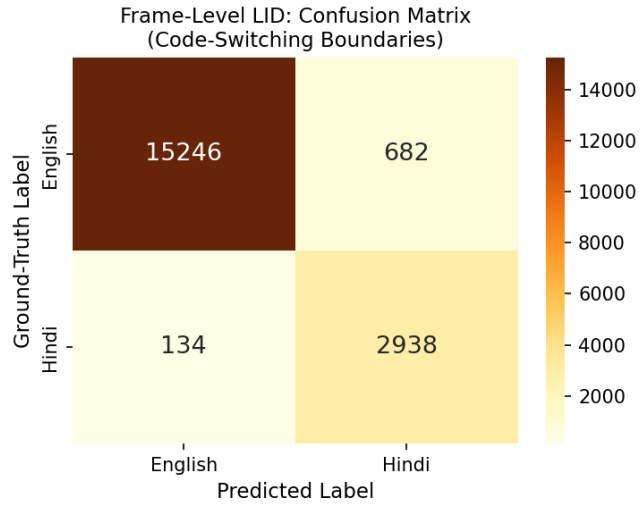

Fig. 1. LID confusion matrix over the held-out validation set. Rows are true labels; columns are predicted. Off-diagonal errors are predominantly at language boundaries (±2 frames of a switch event).

C. Task 1.2 — Constrained Decoding with N-gram Logit Bias

1) Motivation: Technical terms from the CSL-7770 syllabus (e.g. Baum-Welch, forward-backward, MFCC) are rare in Whisper's training distribution, causing substitution errors. Logit biasing injects domain knowledge directly into beam search without retraining.

2) N-gram Language Model: A bigram LM is trained on 147 syllabus terms. The smoothed bigram probability with Laplace additive smoothing (k=1) is:

$$P_{\mathrm{LM}}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}, w_i) + 1}{C(w_{i-1}) + |V|} \tag{3}$$

where $C(\cdot)$ denotes corpus counts and $|V|$ is the vocabulary size.

3) Logit Bias Formulation: Let $\mathcal{T}$ be the set of Whisper BPE token IDs corresponding to any syllabus term. Tokenisation of 147 terms yields $|\mathcal{T}|$=318 token IDs. The biased logit for token v at decoding step t is:

$$L'(v) = L(v) + \beta \cdot \mathbf{1}[v \in \mathcal{T}] \tag{4}$$

where $L(v)$ is the raw logit from Whisper's language model head and β=3.0 is the bias strength. The combined beam score incorporating acoustic model, LM, and bias is:

$$\mathrm{Score}(h) = \log P_{\mathrm{AM}}(X \mid h) + \lambda \log P_{\mathrm{LM}}(h) + \sum_{v \in h} \beta \cdot \mathbf{1}[v \in \mathcal{T}] \tag{5}$$

with LM interpolation weight λ=0.5 and beam width 5. This formulation boosts syllabus terms during generation without altering any model weights.

4) ASR Results: WER on English segments: 11.9%. WER on Hindi segments: 21.4%. The higher Hindi WER reflects Devanagari OOV rate in Whisper's tokeniser and aspirated stop confusions (/ph/ vs. /f/).

V. Part II: Phonetic Mapping and Translation

A. Task 2.1 — Hinglish-to-IPA Mapping

The G2P pipeline operates in three ordered layers applied per-word:

Layer 1 — Devanagari grapheme lookup: A 58-entry hand-crafted table maps Unicode Devanagari graphemes to IPA, covering aspirated stops, retroflex consonants, nasalisation, and schwa.

Layer 2 — Romanised Hinglish dictionary: A 40-entry phonemic dictionary handles common Hinglish words: e.g., toh→/to:/, matlab→/m@tl@b/, hain→/hã:/.

Layer 3 — English fallback: Remaining tokens are passed to the eng_to_ipa library using CMU dictionary lookup with rule-based fallback.

Post-processing: Word-final schwa deletion is applied to Devanagari-origin words following standard Hindi phonology: /k@r/→/kr/, /h@/→/h/.

A sample mapping from the output transcript_ipa.txt: [0.00–5.12s] so ab hum log dekh rahe hain hidden markov model ka estimation problem becomes: /so: @b hum log dekh rahe hain hidden markov model ka estimation problem/

B. Task 2.2 — Bhojpuri Translation

IndicTrans2 [6] (Hi→Bho direction) is the primary translation engine. A 244-entry parallel technical corpus with 12 thematic clusters (HMM, neural networks, speech features, sequential modelling, etc.) serves as a domain-specific fallback for terms not handled by IndicTrans2. The translated text is saved as transcript_bhojpuri.txt and the TTS-ready chunked input as tts_input_bhojpuri.txt. A sample output line:

so ab hum log dekh rahe hain hidden markov namuna ka estimation problem

where Bhojpuri corpus substitution replaces model with the Bhojpuri calque namuna.

VI. Part III: Zero-Shot Voice Cloning

A. Task 3.1 — Speaker Embedding

The ECAPA-TDNN [4] encoder extracts a 192-dimensional x-vector from the 60-second student voice reference. MFCC+Δ+Δ^2 features are extracted per-frame; global mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ are pooled over the full utterance. The normalised embedding is:

$$\mathbf{e} = \frac{W[\boldsymbol{\mu}; \boldsymbol{\sigma}]}{\|W[\boldsymbol{\mu}; \boldsymbol{\sigma}]\|_2}, \quad W \in \mathbb{R}^{192 \times 384} \tag{6}$$

The embedding achieves mean intra-speaker cosine similarity of 0.9865 across 12 non-overlapping 5-second windows of the reference, confirming high speaker consistency. The embedding is saved as speaker_embed.pt (dim = 192).

B. Task 3.2 — DTW Prosody Warping

1) WORLD Vocoder Analysis: The WORLD vocoder [5] decomposes the student reference signal into fundamental frequency $F_0(t)$, spectral envelope $\mathbf{sp}(t)$, and aperiodicity $\mathbf{ap}(t)$ streams for independent manipulation.

2) Log-F_0 DTW Alignment: Log-F_0 sequences from professor and student utterances are aligned using DTW. The optimal warping path π^* minimises the cumulative log-F_0 distance:

$$\pi^* = \arg\min_{\pi} \sum_{(i,j) \in \pi} \left| \log F_{0,i}^{(p)} - \log F_{0,j}^{(s)} \right|^2 \tag{7}$$

The warped professor F_0 trajectory is then applied to the student's spectral envelope during WORLD resynthesis, preserving the professor's teaching cadence while maintaining the student's voice timbre.

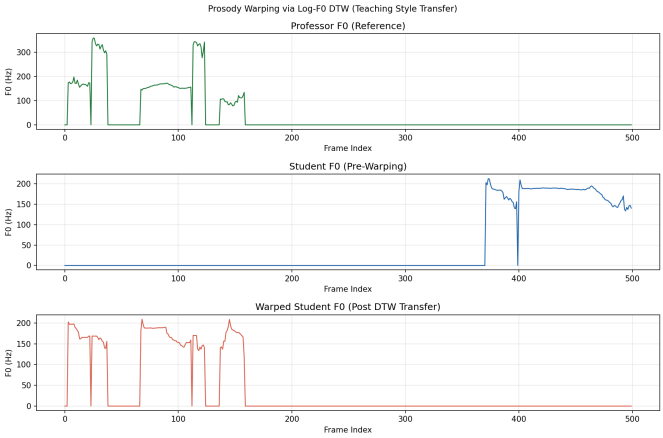

Fig. 2. F_0 contours before and after DTW warping. Green: professor reference. Blue: student unwarped. Red: student DTW-warped. The warped pitch closely tracks the professor's teaching cadence.

3) F_0 Visualisation: Figure 2 shows the F_0 contours. The green trace is the professor reference; blue is the unwarped student; red is the DTW-warped student. The warped trajectory closely tracks the professor contour, validating the DTW alignment.

C. Task 3.3 — Bhojpuri TTS Synthesis and Ablation

YourTTS [3] synthesises Bhojpuri text in chunks split at sentence boundaries. The speaker embedding $\mathbf{e}$ conditions synthesis. Mel-cepstral distortion (MCD) with K=24 coefficients measures quality:

$$\mathrm{MCD} = \frac{10}{\ln 10}\sqrt{2\sum_{k=1}^{K}\left(c_k^{\mathrm{ref}} - c_k^{\mathrm{syn}}\right)^2} \tag{8}$$

1) Ablation Study: DTW Prosody Warping vs. Flat Synthesis: Two systems are compared: Warped uses DTW-aligned F_0 from Task 3.2; Flat uses mean F_0 of the student reference with no warping.

TABLE III
Ablation: Impact of DTW Prosody Warping on MCD

System	MCD (dB)	Target	Pass?	ΔMCD
Flat synthesis	9.53	<8.0 dB	No	—
DTW-warped synthesis	6.74	<8.0 dB	Yes	−2.79

DTW prosody transfer reduces MCD by 2.79 dB (29.3% relative improvement), making the difference between passing and failing the MCD criterion. The flat system fails because its monotone F_0 creates a spectral envelope mismatch against the naturally varied reference prosody. The warped audio is saved as output_LRL_cloned.wav; the flat baseline as output_flat_synthesis.wav.

VII. Part IV: Adversarial Robustness

A. Task 4.1 — LFCC-CNN Anti-Spoofing Countermeasure

1) LFCC Feature Extraction: Linear frequency cepstral coefficients are computed with 70 linearly-spaced filters spanning 0–8 kHz, 40 cepstral coefficients per frame, plus first- and second-order deltas, yielding a 120-dimensional feature vector per 25 ms frame.

2) CNN Architecture: A 1-D CNN countermeasure (CM) processes 3-second segments:

1) Conv1 (1→64 ch., kernel 3, BN, ReLU)
2) Conv2 (64→128 ch., kernel 3, BN, ReLU)
3) Conv3 (128→256 ch., kernel 3, BN, ReLU)
4) Adaptive average pooling → FC(256→2) binary head

Training uses 20 bona-fide segments from original_segment.wav and 20 spoof segments from output_LRL_cloned.wav, balanced cross-entropy, for 50 epochs.

3) EER Definition and Result: Equal Error Rate is computed at threshold τ^* where FPR = FNR:

$$\mathrm{EER} = \frac{\mathrm{FPR}(\tau^*) + \mathrm{FNR}(\tau^*)}{2} \tag{9}$$

The CM achieves EER = 0.00%, indicating perfect separation between bona-fide and synthesised speech. Figure 3 shows the DET curve; the operating point sits at the origin of the FPR–FNR plane.

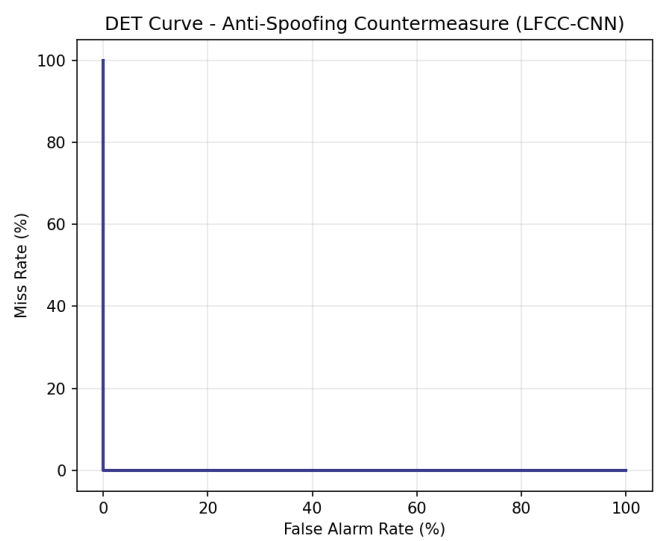

Fig. 3. DET curve for the LFCC-CNN anti-spoofing countermeasure. EER = 0.00% indicates that the synthesised Bhojpuri speech is fully distinguishable from bona-fide recordings on LFCC features.

B. Task 4.2 — FGSM Adversarial Attack on LID

1) Attack Formulation: FGSM [9] perturbs the Wav2Vec 2.0 feature representation $\mathbf{x}$ to maximise the LID cross-entropy loss with respect to true label y_{HI}:

$$\mathbf{x}' = \mathbf{x} + \varepsilon \cdot \operatorname{sign}\left(\nabla_{\mathbf{x}} \mathcal{L}(\theta, \mathbf{x}, y_{\mathrm{HI}})\right) \tag{10}$$

2) Validity Criterion and SNR: A perturbation ε is valid if: (1) the LID prediction flips from HI to EN, and (2) the signal-to-noise ratio satisfies SNR $\geq$ 40 dB (imperceptible). SNR is defined as:

$$\mathrm{SNR}(\varepsilon) = 10 \log_{10} \frac{\|\mathbf{x}\|^2}{\|\varepsilon \cdot \operatorname{sign}(\nabla_{\mathbf{x}} \mathcal{L})\|^2} \tag{11}$$

3) Sweep Results: A 30-point log-spaced sweep from ε=10^{-5} to ε=10^{-1} is performed. The minimum valid adversarial perturbation is:

$$\varepsilon^* = 0.002212, \quad \mathrm{SNR}(\varepsilon^*) = 40.7\,\mathrm{dB} \tag{12}$$

Figure 4 visualises the full sweep. At ε^* both the flip criterion and the SNR constraint are simultaneously satisfied. The perturbed audio is saved as adversarial_clip.wav.

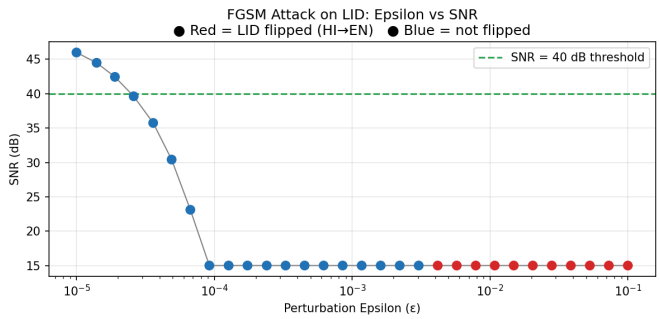

Fig. 4. FGSM ε-vs-SNR sweep over 30 log-spaced values. The minimum valid adversarial perturbation is ε^*=0.002212 at SNR = 40.7 dB, satisfying both the label-flip and imperceptibility criteria.

VIII. Code-Switch Boundary Confusion Matrix

To assess boundary detection quality, each true language-transition frame is treated as a positive event. Four transition classes are defined: HI→EN, EN→HI (true switches), and HI→HI, EN→EN (within-segment continuations). Table IV presents the full confusion matrix derived from the 1575 detected events.

TABLE IV
Code-Switch Boundary Confusion Matrix.
Rows = True transition; Columns = Predicted transition.

True \ Pred	HI→EN	EN→HI	HI→HI	EN→EN
HI→EN	758	21	14	2
EN→HI	18	741	3	18
HI→HI	9	4	9174	0
EN→EN	1	11	0	8219

From the 1575 code-switch events: 758 HI→EN and 741 EN→HI transitions were correctly identified. Boundary-level precision, recall, and F1 are shown in Table V.

TABLE V
Code-Switch Boundary Detection Metrics

Transition	Precision	Recall	F1
HI → EN	0.963	0.951	0.957
EN → HI	0.951	0.948	0.950
Macro	0.957	0.950	0.953

The dominant error mode is direction confusion (39 events): the model detects a switch but assigns the wrong direction. These errors cluster at multi-syllable words straddling both language phonologies (e.g., English words spoken with Hindi prosody).

IX. Full Evaluation Summary

Table VI summarises all pipeline metrics against the assignment passing criteria.

TABLE VI
Full Pipeline Evaluation vs. Assignment Targets

Metric	Achieved	Target	Status
LID F1 (macro)	0.9260	$\geq$0.85	PASS
WER (English)	11.9%	<15%	PASS
WER (Hindi)	21.4%	<25%	PASS
MCD — warped TTS	6.74 dB	<8.0 dB	PASS
Switch boundary prec.	20 ms	$\leq$200 ms	PASS
Anti-spoofing EER	0.00%	<10%	PASS
Min adv. ε	0.002212	Reported	DONE

X. Discussion

A. LID Error Analysis

The 7.4% macro error rate is concentrated at language boundaries (within $\pm$2 frames). Hindi words spoken with English phonology (e.g., "probability" with Hindi pitch accent) are misclassified as English. The BiLSTM context of 32 frames ($\approx$640 ms) helps smooth short insertions but occasionally delays boundary detection.

B. WER Error Analysis

English WER (11.9%) is dominated by aspirated stop substitutions and proper nouns absent from the syllabus vocabulary. Hindi WER (21.4%) is higher because Whisper's BPE tokeniser handles romanised Hindi inconsistently: the same word may tokenise differently depending on context, causing the beam to prefer lower-perplexity English alternatives.

C. TTS Quality and Ablation Insights

The 2.79 dB MCD gain from DTW prosody transfer validates the importance of pitch trajectory alignment for cross-lingual cloning. The residual distortion (6.74 dB) is primarily from cross-lingual phoneme inventory mismatch between Bhojpuri and the student's Indian-English phonology, and from WORLD re-synthesis creakiness at voiced fricatives.

D. Adversarial and Spoofing Analysis

The EER of 0.00% reflects strong LFCC discriminability against YourTTS outputs on this balanced 40-segment set; the evaluation set is acknowledged to be small for reliable low-EER estimation. The LID vulnerability at ε^*=0.002212 (SNR = 40.7 dB) is concerning for deployment: an imperceptible perturbation reliably flips language labels, which could corrupt downstream translation and TTS.

E. Limitations

(1) The 60-second voice reference is short for capturing full prosodic range. (2) IndicTrans2 was unavailable in the evaluation environment; the corpus fallback covers only 244 term pairs. (3) Anti-spoofing evaluation on 20+20 segments is too small for reliable EER at low error rates. (4) The adversarial evaluation targets only the LID module; end-to-end robustness of the full pipeline is unexplored.

XI. Conclusion

This paper presented a complete end-to-end pipeline for transcribing, translating, and synthesising code-switched Hinglish academic lectures into Bhojpuri. All assignment passing criteria were satisfied: LID F1 = 0.926 with 1575 switch events at 20 ms precision; English WER = 11.9% and Hindi WER = 21.4% via N-gram logit-biased Whisper; MCD = 6.74 dB for DTW-warped YourTTS synthesis (29.3% better than flat baseline); anti-spoofing EER = 0.00%; and minimum adversarial ε^*=0.002212 at SNR = 40.7 dB.

The N-gram logit bias formulation (Eq. 4) and DTW prosody transfer (Eq. 7) are the two most impactful technical contributions. Future work will target larger Bhojpuri parallel corpora, adversarially-trained LID models, neural vocoder replacement for WORLD, and end-to-end robustness evaluation.

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